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Nested Hole Saw Set with Quick Connect Arbor and Tube Storage Case

Abstract

A hole saw system includes a first hole saw, a second hole saw, and an arbor. The first hole saw is capable of being nested within the second hole saw. Each of the hole saws includes a mount wall having a first engagement feature configured to receive a first drive feature and a second engagement feature configured to receive a second drive feature that is different from the first drive feature. The arbor includes the first drive feature and the second drive feature and is configured to selectively mount each of the hole saws in a single saw arrangement and to selectively mount both of the hole saws in a nested saw arrangement. In the single saw arrangement, the arbor mounts a single one of the hole saws with the first drive feature engaged in the first engagement feature and the second drive feature not engaged with the second engagement feature. In the nested saw arrangement, the arbor mounts both of the hole saws with the first hole saw nested within the second hole saw and the first drive feature engaged in the first engagement feature of the first hole saw and the second drive feature engaged in the second engagement feature of the second hole saw.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to provisional U.S. Patent Application No. 63/558,927 filed Feb. 28, 2024, the entire disclosure of which is incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

[0002] The present disclosure relates to tools, and more particularly, to a cup-shaped hole saws that are mounted on an arbor or mandrel for cutting circular holes through a work piece, such a door, wall, floor, or stud, and more specifically to such tools wherein a cup-shaped hole saw can be quickly and easily removed from and installed onto an arbor. While such tools are known and work well for their intended purpose, there is always room for improvement.

BRIEF SUMMARY OF THE DISCLOSURE

[0003] In accordance with one aspect of this disclosure, a hole saw system is provided and includes a first hole saw, a second hole saw, and an arbor. The first hole saw is capable of being nested within the second hole saw. Each of the hole saws is sized to cut a different diameter hole than the other hole saw. Each of the hole saws has a cylindrical shaped cutting wall extending upward from a mount wall, with the mount wall including a first engagement feature configured to receive a first drive feature and a second engagement feature configured to receive a second drive feature that is different from the first drive feature. The arbor includes the first drive feature and the second drive feature and is configured to selectively mount each of the hole saws in a single saw arrangement to perform a single hole cut and to selectively mount both of the hole saws in a nested saw arrangement with the first hole saw nested within the second hole saw to perform a double hole cut. In the single saw arrangement, the arbor mounts a single one of the hole saws with the first drive feature engaged in the first engagement feature and the second drive feature not engaged with the second engagement feature. In the nested saw arrangement, the arbor mounts both of the hole saws with the first hole saw nested within the second hole saw and the first drive feature engaged in the first engagement feature of the first hole saw and the second drive feature engaged in the second engagement feature of the second hole saw.

[0004] As one aspect, the mount wall of each of the hole saws includes a plurality of the first engagement feature and a plurality of second engagement features, with the first engagement features being equally circumferentially spaced from each other and the second engagement features being equally circumferentially spaced from each other.

[0005] In one aspect, the first drive feature extends above the second drive feature.

[0006] According to one aspect, the first drive feature is a pin and each of the first engagement features is a bore extending into the mount wall and shaped to receive the pin. In a further aspect the pin is a cylindrical shaped pin and the bore is shaped to conform to the cylindrical shape of the pin.

[0007] As one aspect, the second drive feature is defined by at least one rib and each of the second engagement features is an opening in the mount wall shaped to receive the at least one rib. In a further aspect, the at least one rib defines an upwardly facing surface that engages against a

downwardly facing surface of the mount wall of each of the hole saws with the first drive feature engage in the first engagement feature of the mount wall. As another aspect, the at least one rib extends from the pin, and the opening in each of the mount walls is shaped to receive the pin. In a further aspect, the at least one rib is two ribs extending from opposite sides of the pin.

[0008] According to one aspect, each of the hole saws include an additional first engagement feature and an additional second engagement feature, and the arbor includes an additional first drive feature and an additional second drive feature.

[0009] In one aspect, the arbor further includes a locking member that engages each of the hole saws to retain the hole saw on the arbor with the first drive feature engaged in the first engagement feature of the hole saw.

[0010] According to one aspect, the locking member defines a shoulder that engages against a surface of the mount wall of each of the hole saws to retain the hole saw on the arbor with the first drive feature engaged in the first engagement feature of the hole saw.

[0011] As one aspect, the locking member is mounted to the arbor to move between a loading position wherein each of the hole saws can be loaded onto the arbor to engage the first drive feature in the first engagement feature and a locking position wherein the locking member engages the hole saw to retain the hole saw on the arbor with the first drive feature engaged in the first engagement feature.

[0012] According to one aspect, the arbor further includes an actuator that can be operated by a user to move the locking member from the locking position to the loading position.

[0013] As one aspect, the arbor includes an additional locking member that engages each of the hole saws to retain the hole saw on the arbor with the first drive feature engaged in the first engagement feature of the hole saw, the locking members being located opposite of each other on the arbor.

[0014] In one aspect, the hole saw system further includes a third hole saw that can be nested within the first hole saw, the third hole saw having a cylindrical shaped cutting wall extending upward from a mount wall.

[0015] According to one aspect, the third hole saw includes a third engagement feature formed in the cylindrical cutting wall of the first hole saw, the third engagement feature shaped to receive a drive portion of the locking member.

[0016] As one aspect, the cylindrical cutting wall of the third hole saw is spaced radially inward from the second drive feature with drive portion of the locking member engaged in the third engagement feature.

[0017] In one aspect, the mount wall of at least one of the first and second hole saws have a central opening sized to allow the cylindrical cutting wall of the third hole saw to pass through the mounting wall to allow the third hole saw to be mounted on the arbor nested within the at least one of the first and second hole saws.

Description

BRIEF SUMMARY OF THE SEVERAL VIEWS OF THE DRAWINGS

[0018] FIG. 1 perspective view from above of a hole saw system according to this disclosure;

[0019] FIG. 2 is perspective view from above of a hole saw of the system of FIG. 1;

[0020] FIG. 3 is view similar to FIG. 1 but showing the system with the hole saw of FIG. 2 removed;

[0021] FIG. 4 is a perspective view from above of an arbor of the system of FIG. 1;

[0022] FIG. 5 is a view similar to FIG. 1, but showing the system in a single saw arrangement with a single hole saw mounted on the arbor;

[0023] FIG. 6 is a section view taken along line 6-6 in FIG. 5;

[0024] FIG. 7 is a broken section view taken along line 7-7 in FIG. 5;
[0025] FIG. 8 is a view similar to FIG. 5, but taken from a different angular orientation;
[0026] FIG. 9 is a top view of the system of FIG. 1;
[0027] FIG. 10 is a perspective from above of the system of FIG. 1;
[0028] FIG. 11 is a section view taken along line 11-11 in FIG. 9;
[0029] FIG. 12 is another perspective view from above of the system of FIG. 1;
[0030] FIG. 13 is a section perspective view taken from line 13-13 in FIG. 12;
[0031] FIG. 14 is a somewhat diagrammatic longitudinal cross section of the arbor of FIG. 4, showing locking members of the arbor in an unlocked or loading position;
[0032] FIG. 15 is similar to FIG. 14 but showing the locking members in a locking position;
[0033] FIG. 16 is a perspective view of the arbor of FIGS. 4, 15, and 16 with another size of hole saw mounted thereon;
[0034] FIG. 17 is a section view taken from line 17-17 in FIG. 16;
[0035] FIG. 18 is a top view of the hole saw of FIGS. 16 and 17 mounted in a nested saw arrangement on the arbor with one of the hole saws of FIG. 1;
[0036] FIG. 19 is a section perspective view taken from line 19-19 in FIG. 18;
[0037] FIG. 20 is a perspective view from above of the arbor of FIG. 4 with yet another size of hole saw carried thereon;
[0038] FIG. 21 is a longitudinal cross section of the arbor of FIG. 4 with the hole saw of FIG. 20 mounted in a nested saw arrangement on the arbor with one of the hole saws of FIG. 1;
[0039] FIG. 24 is a side elevation view of a storage case for the hole saw system according to this disclosure, with the hole saw system storage mounted therein and some of the case components formed from transparent material,
[0040] FIG. 25 is a view similar to FIG. 24 but showing the system 10 stored in a different configuration within the storage case of FIG. 24;
[0041] FIG. 22 is another embodiment of the storage case according to this disclosure wherein all of the case components are formed from non-transparent materials;
[0042] FIG. 23 is a longitudinal cross section of the storage case of FIG. 22 showing the hole saw system storage mounted therein;
[0043] FIG. 26 is an exploded view of the storage case of FIGS. 24 and 25, with the hole saw system stored in the configuration shown in FIG. 24;
[0044] FIG. 27 is a view similar to FIG. 26 but with the hole saw system removed;
[0045] FIG. 28 is a perspective view from above of a top case member of the storage case according to this disclosure;
[0046] FIG. 29 is a section perspective view of the top case member of FIG. 28;
[0047] FIG. 30 is a perspective view from above of a center case member of the storage case according to this disclosure;
[0048] FIG. 31 is a section perspective view of the center case member of FIG. 30;
[0049] FIG. 32 is a perspective view from above of a base member of the storage case according to this disclosure; and
[0050] FIG. 33 is a section perspective view of the base member of FIG. 32.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0051] With reference to FIG. 1, a hole saw system 10 is shown and includes a first hole saw 12, a second hole saw 14, and an arbor 16. As shown, the first hole saw 12 is capable of being nested within the second hole saw 14, with each of the hole saws 12 and 14 being sized to cut a different diameter hole. Each of the hole saws 12 and 14 includes a cylindrical shaped cutting wall 18 extending upward from a mount wall 20. In the illustrated embodiments, each of the mount walls 20 includes a first pair of first engagement features 22 and a pair of second engagement features 24, as best seen in FIG. 2 and FIG. 3. In the illustrated embodiment, each of the mount walls 20 further includes a second pair of first engagement features 22, each associated with one of the second

engagement features **24**. As best seen in FIG. **4**, in the illustrated embodiments, the arbor **16** includes a pair of first drive features **26** and a pair of second drive features **28**. As best seen in FIGS. **1** and **2**, each of the first engagement features **22** is configured to receive one of the first drive features **26** and each of the second engagement features **24** is configured to receive one of the second drive features **28**.

[0052] As best seen in FIGS. **5-8**, the arbor **16** is configured to selectively mount each of the hole saws **12** and **14** in a single saw arrangement to perform a single hole cut. In the single saw arrangement, the arbor **16** mounts a single one of the hole saws **12** and **14** with the first drive features **26** engaged in a pair of the first engagement features **22** and the second drive features **28** not engaged with the second engagement features **24**. As best seen in FIGS. **1**, **3**, **9**, and **10**, the arbor **16** is also configured to selectively mount both of the hole saws **12** and **14** in a nested saw arrangement, with the first hole saw **12** nested within the second hole saw **14** to perform a double hole cut. In the nested saw arrangement, the arbor **16** mounts the first hole saw **12** nested within the second hole saw **14**, with the first drive feature **26** engaged in a pair of the first engagement feature **22** of the first hole saw **12** and the second drive feature **28** engaged in the second engagement feature **24** of the second hole saw **14** (best seen in FIG. **3**).

[0053] In the orientation shown in the illustrated embodiments, the first drive features **26** extend above the second drive feature **28**. In the illustrated embodiments, the first drive feature **26** is a cylindrical shaped pin **26** and each of the first engagement features **22** is a cylindrical shaped bore **22** extending through the mount wall **20** and sized to receive the pin **26**. As best seen in FIG. **4**, in the illustrated embodiment, each of the second drive features **28** is defined by a pair of arch shaped ribs **30** extending from opposite sides of an associated one of the pins **26** and each of the second engagement features **24** is a through opening **24** in the mount wall shaped to conform to receive the ribs **30**.

[0054] As best seen in FIG. **6**, in at least some of the embodiments, each of the ribs **30** defines an upwardly facing surface **32** that abuts against a downwardly facing surface **34** of the mount wall **20** of each of the hole saws **12** and **14** with the first drive feature **26** engage in the first engagement feature **22** that are not associated with the second engagement features **24**.

[0055] It should be appreciated that the arbor **16** includes mount structure **36** for drivingly mounting a drill bit **37**, as is common for the arbors of many known hole saw systems.

[0056] As best seen in FIGS. **7** and **11**, in the illustrated embodiment, the arbor **14** further includes a pair of locking members **40** that can selectively engage each of the hole saws **12** and **14** to retain the hole saw on the arbor **16** with the first drive features **26** engaged in a pair of the first engagement feature **22**. In this regard, in the illustrated embodiment, each of the locking members **40** defines a shoulder **44** that engages against a surface **46** of the mount wall **20** of each of the hole saws **12** and **14** to retain the hole saw on the arbor **16** with the first drive features **26** engaged in a pair of the first engagement feature **22**.

[0057] As best seen in FIGS. **14** and **15**, each of the locking members **40** is mounted in the arbor **16** to move between an unlocked or loading position (shown in FIGS. **1**, **3**, **8**, **9**, **12-14**, and **21**) wherein each of the hole saws **12** and **14** can be loaded onto the arbor **16** to engage the first drive features **26** in a pair of the first engagement features **22**, and a locking position (shown in FIGS. **5**, **7**, **10**, **11**, **15**, **16**, **18**, and **19**) wherein the locking members **40** engages the hole saw **12**, **14** to retain the hole saw **12**, **14** on the arbor **16** with the first drive features **26** engaged in a pair of the first engagement features **22**. In the illustrated embodiment, the arbor **16** further includes an actuator unit **48** that can be operated by a user to move the locking members **40** from the locking position to the loading position. While preferred embodiments of the actuator unit **48** are shown in the Figs., it should be understood that any suitable mechanism for engaging and disengaging the locking members to lock and unlock hole saws to the arbor **16** can be utilized.

[0058] As best seen in FIGS. **16-19**, in the illustrated embodiment, the system **10** includes a third hole saw **50** that can be nested within either the first hole saw **12** or the second hole saw **14**. As

with the other hole saws of the system, the third hole saw **50** has a cylindrical shaped cutting wall **52** extending upward from a mount wall **54**. As can be seen in FIGS. **16-19**, the third hole saw is configured to cut a much smaller diameter hole than the hole saws **12** and **14**, and accordingly the cutting wall **52** is of a much smaller diameter. This smaller size will not allow for the third hole saw **50** to incorporate the first and second engagement features **22** and **24** of the other hole saws **12** and **14**. Accordingly, the third hole saw **50** a pair of third engagement features **56** that are formed in the cylindrical cutting wall **52**, with each of the features **56** being a recess **56** shaped to receive a drive portion **58** of a corresponding one of the locking members **40**. As can be seen in FIGS. **16-19**, when mounted on the arbor **16** with drive portions **58** engaged in the recesses **56**, the cylindrical cutting wall **52** of the third hole saw **50** is spaced radially inward from the first and second drive features **24** and **26**. As best seen in FIGS. **18** and **19**, the mount wall **20** of each of the first and second hole saws **12** and **14** has a central through opening or bore **62** sized to allow the cylindrical cutting wall **52** to pass through the mounting wall **20** to allow the third hole saw **50** to be mounted on the arbor **16** nested within the either one of the first and second hole saws **12** and **14**.

[0059] As shown in FIGS. **20**, the system **10** can include hole saws of other sizes, such as the hole saw **64**, that incorporate first and second engagement features **22** and **24** that are compatible with the first and second drive features **26** and **28** on the arbor **16** to allow further hole size options when the additional hole saws are mounted on the arbor in either the single saw arrangement or the nested saw arrangement. As best seen in FIGS. **23-26**, the system can include a large number of such additional hole saws.

[0060] As shown in FIGS. **23-33**, a tube-shaped storage case **70** can be included to store the system **10** and to allow for convenient use of the component of the system **10**. As best seen in FIG. **27**, in the illustrated embodiments, the case **70** includes a base **72**, center case member **74**, and a top case member **76**. The center case member **74** includes screw threads **78** at its lower end to engage with mating screw threads **80** on the base **72** and further includes screw threads **82** at its upper end to engage with mating screw threads **84** in the lower end of the top case member **76**. The base **72** is configured to carry a plurality of the hole saws **12**, **14**, **50**, **64**, etc. for storage. The Center case member **74** is configured to carry the arbor **16** with one or two of the hole saws **12**, **14**, **50**, **64**, etc. mounted on the arbor **16** in either the single saw or nested saw arrangements.

Claims

1. A hole saw system comprising: a first hole saw and a second hole saw, the first hole saw capable of being nested within the second hole saw, each of the hole saws sized to cut a different diameter hole than the other hole saw, each of the hole saws having a cylindrical shaped cutting wall extending upward from a mount wall with the mount wall including a first engagement feature configured to receive a first drive feature and a second engagement feature configured to receive a second drive feature that is different from the first drive feature; and an arbor comprising the first drive feature and the second drive feature and configured to selectively mount each of the hole saws in a single saw arrangement to perform a single hole cut and to selectively mount both of the hole saws in a nested saw arrangement with the first hole saw nested within the second hole saw to perform a double hole cut, wherein: in the single saw arrangement, the arbor mounts a single one of the hole saws with the first drive feature engaged in the first engagement feature and the second drive feature not engaged with the second engagement feature, and in the nested saw arrangement, the arbor mounts both of the hole saws with the first hole saw nested within the second hole saw and the first drive feature engaged in the first engagement feature of the first hole saw and the second drive feature engaged in the second engagement feature of the second hole saw.

2. The hole saw system of claim 1 wherein the mount wall of each of the hole saws includes a plurality of the first engagement feature and a plurality of second engagement features, with the first engagement features being equally circumferentially spaced from each other and the second

engagement features being equally circumferentially spaced from each other.

3. The hole saw system of claim 1 wherein the first drive feature extends above the second drive feature.

4. The hole saw system of claim 1 wherein the first drive feature is a pin and each of the first engagement features is a bore extending into the mount wall and shaped to receive the pin.

5. The hole saw system of claim 4 wherein the pin is a cylindrical shaped pin and the bore is shaped to conform to the cylindrical shape of the pin.

6. The hole saw system of claim 4 wherein the second drive feature is defined by at least one rib and each of the second engagement features is an opening in the mount wall shaped to receive the at least one rib.

7. The hole saw system of claim 6 wherein the at least one rib defines an upwardly facing surface that engages against a downwardly facing surface of the mount wall of each of the hole saws with the first drive feature engage in the first engagement feature of the mount wall.

8. The hole saw system of claim 6 wherein the at least one rib extends from the pin, and the opening in each of the mount walls is shaped to receive the pin.

9. The hole saw system of claim 8 wherein the at least one rib is two ribs extending from opposite sides of the pin.

10. The hole saw system of claim 6 is the at least one rib is an arch shaped rib.

11. The hole saw system of claim 1 wherein each of the hole saws include an additional first engagement feature and an additional second engagement feature, and the arbor comprises an additional first drive feature and an additional second drive feature.

12. The hole saw system of claim 1 wherein the arbor further comprises a locking member that engages each of the hole saws to retain the hole saw on the arbor with the first drive feature engaged in the first engagement feature of the hole saw.

13. The hole saw system of claim 12 wherein the locking member defines a shoulder that engages against a surface of the mount wall of each of the hole saws to retain the hole saw on the arbor with the first drive feature engaged in the first engagement feature of the hole saw.

14. The hole saw system of claim 12 wherein the locking member is mounted to the arbor to move between a loading position wherein each of the hole saws can be loaded onto the arbor to engage the first drive feature in the first engagement feature and a locking position wherein the locking member engages the hole saw to retain the hole saw on the arbor with the first drive feature engaged in the first engagement feature.

15. The hole saw system of claim 14 wherein the arbor further comprising an actuator that can be operated by a user to move the locking member from the locking position to the loading position.

16. The hole saw system of claim 12 wherein the arbor comprises an additional locking member that engages each of the hole saws to retain the hole saw on the arbor with the first drive feature engaged in the first engagement feature of the hole saw, the locking members being located opposite of each other on the arbor.

17. The hole saw system of claim 12 further comprising a third hole saw that can be nested within the first hole saw, the third hole saw having a cylindrical shaped cutting wall extending upward from a mount wall.

18. The hole saw system of claim 17 wherein the third hole saw comprises a third engagement feature formed in the cylindrical cutting wall of the first hole saw, the third engagement feature shaped to receive a drive portion of the locking member.

19. The hole saw system of claim 18 wherein the cylindrical cutting wall of the third hole saw is spaced radially inward from the second drive feature with drive portion of the locking member engaged in the third engagement feature.

20. The hole saw system of claim 19 wherein the mount wall of at least one of the first and second hole saws have a central opening sized to allow the cylindrical cutting wall of the third hole saw to

pass through the mounting wall to allow the third hole saw to be mounted on the arbor nested within the at least one of the first and second hole saws.
